LOVELY PROFESSIONAL UNIVERSITY · INT335 DESIGN THINKING CAPSTONE SHOWCASE

RETURNLOOP LPU: RETHINKING LOST & FOUND ON A 600-ACRE CAMPUS

Project ID: P04 | **Section:** K212520 | **Team:** 15 | **Mentor:** Dr. Sarabjit Kaur | **Live URL:** https://returlooptesting-khwd.vercel.app/

1. EMPATHY & PROBLEM SPACE

The 600-Acre Campus Challenge:
- Over 35,000 students across 60+ academic blocks and 40 hostels.
- 10-minute lecture transitions trigger high-frequency item loss.
- Cardboard lost property boxes sit uncatalogued at gate desks.
- Informal WhatsApp groups invite fake claims, extortion, and phone spam.

Primary User Personas:
- **Aarav Sharma (20, CSE Hosteller · BH-4):** Lost Casio fx-991EX calculator before midterms; got 4 fake claims on social media.
- **Priya Patel (21, Biotech Day Scholar):** Commutes to Jalandhar; found keys at UniMall but had only 5 mins before bus departed.

Key Field Evidence (8 Interviews & AEIOU):
- **Rohan Verma:** "Someone tried to claim my calculator to get a free device."
- **Simranjeet Kaur:** "Lost ID card; locked out of mess and library for 4 days."
- **Officer Gurpreet:** "6 unindexed black umbrellas under desk; cannot verify owner."

Point of View (POV):
Students need an authenticated, privacy-preserving recovery channel without exposing personal phone numbers because informal social media invites fraud and harassment.

2. RETURNLOOP PROTOTYPE & ARCHITECTURE

System Innovations & Core Differentiators:
1. **5-Trait Similarity Matching:** Auto-compares category, brand, model, color, and location in < 20ms.
2. **Masked Privacy Listings:** Public items hide contact details; zero phone numbers leaked.
3. **Secret Identifier Verification:** Hidden serials/engravings only seen by security staff.
4. **7 Indian Languages:** Hindi, Punjabi, Telugu, Tamil, Bengali, Marathi, English.
5. **Campus Admin Desk:** Digital audit trail logs item verifications and handovers.

Live Working Deployment:
- Live Progressive Web App: **https://returlooptesting-khwd.vercel.app/**
- Full-Stack: React 18 + Vite + Tailwind CSS + Python Flask REST API
- Runs on student mobile browsers with sub-second page loads at ■0 cost.

User Handover Workflow:
Loser logs item + secret proof ➔ Finder logs item in 45s ➔ Engine auto-matches ➔ Owner presents claim slip at Security Desk ➔ Guard verifies secret proof & hands over.

3. TESTING, ITERATIONS & TEAM ROLES

Quantitative Usability Testing (6 Campus Participants):
- 100% task completion success across all 6 participants.
- Average report time: 38 seconds (down from 15 minutes manual search).

The 4 Core Engineering Fixes Implemented:
1. **Mobile WebKit Fix:** React createPortal solves iPhone safe-area modal clipping.
2. **Staff Accessibility:** Added Punjabi/Hindi UI so guards verify items in 15 seconds.
3. **Secret Proof Lock:** Added visual tooltips preventing public disclosure of serial numbers.
4. **Vite Build Optimization:** Resolved import analysis with unified TranslatedText.

Measured Campus Impact:
- **Recovery Time:** Cut from 4 days to < 2 hours.
- **Privacy Leaks:** 100% reduction in phone number broadcasts.
- **Impostor Claims:** 0% false claims processed.
- **Cost:** ■0 capital expenditure.

Team 15 Role Division on Board:
- **G. Bhuvanasai (Lead · 12508429):** Architecture, Prototyping & Live Demo.
- **Abin Raj MR (Research · 12509293):** 8 Interviews, AEIOU & Personas.
- **Adarsh (Testing · 12517821):** Usability Testing, Metrics & 4 Fixes.
- **Piyush Mehra (Presentation · 12509335):** 32 Ideas Triage, NABC & Showcase.

< 2 HOURS	100% PROTECTED	0% IMPOSTOR FRAUD	7 LANGUAGES	■0 BUDGET
Average Item Recovery Time	Zero Public Phone Number Leaks	Secret Identifier Verification	Hindi, Punjabi, Telugu, etc.	Pure Human-Centered Design